

Haptic Sim-Chair-Based Teleoperation System for Autonomous Vehicle Platforms

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Abstract

Remote operation of ground vehicles presents fundamental challenges in situational awareness. An operator who is not physically present loses sensory feedback for precise, confident driving, including peripheral vision and the perception of acceleration. Conventional teleoperation systems provide only a flat video feed, limiting use in hazardous environments and autonomy safety intervention.

The Autonomous Vehicles Laboratory (AVL) at California State Polytechnic University, Pomona (CPP) has developed a bidirectional haptic teleoperation system that restores this missing feedback. The test platform carries three ZED X stereo cameras and an Xsens MTi-68oG IMU on an NVIDIA Jetson Orin. The operator sits in a D-BOX motion chair with a Thrustmaster wheel and pedals, facing three monitors that recreate a driver's field of view. Steering and pedal inputs go to the vehicle in real time, while measured accelerations drive the chair so the operator feels the forces on the vehicle.

All communication runs over a Tailscale VPN rather than line-of-sight radio, so the operator and vehicle can be any distance apart. This poster presents the system architecture, low-latency video and control over the internet, and applications of immersive teleoperation to autonomous vehicle development.

Introduction

Remote drivers lose the cues that make driving intuitive: peripheral vision, body motion, and the feel of acceleration. This project restores those senses through a bidirectional haptic teleoperation system. Three coupled channels run over Tailscale VPN & in this case control MARVIN, AVL's autonomous car.

- **Vision:** panoramic live video from three mounted ZED X cameras
- **Control:** low-latency wheel and pedal commands to differential drive
- **Haptics:** IMU accelerations mapped through a washout filter into D-BOX chair motion



Figure 1. MARVIN, test vehicle for this system.

Objectives

- **Latency:** median glass-to-glass camera delay under 150 ms
- **Control feel:** smooth remote driving comparable to a real life set
- **Motion feedback:** real accel & brake felt via the D-BOX chair
- **Integration:** cameras, controls, and haptic chair running together

Methodology

Three parallel pipelines connect the Windows sim-chair station to the vehicle's NVIDIA Jetson Orin (Ubuntu 22.04, ROS 2 Humble) over a Tailscale VPN.

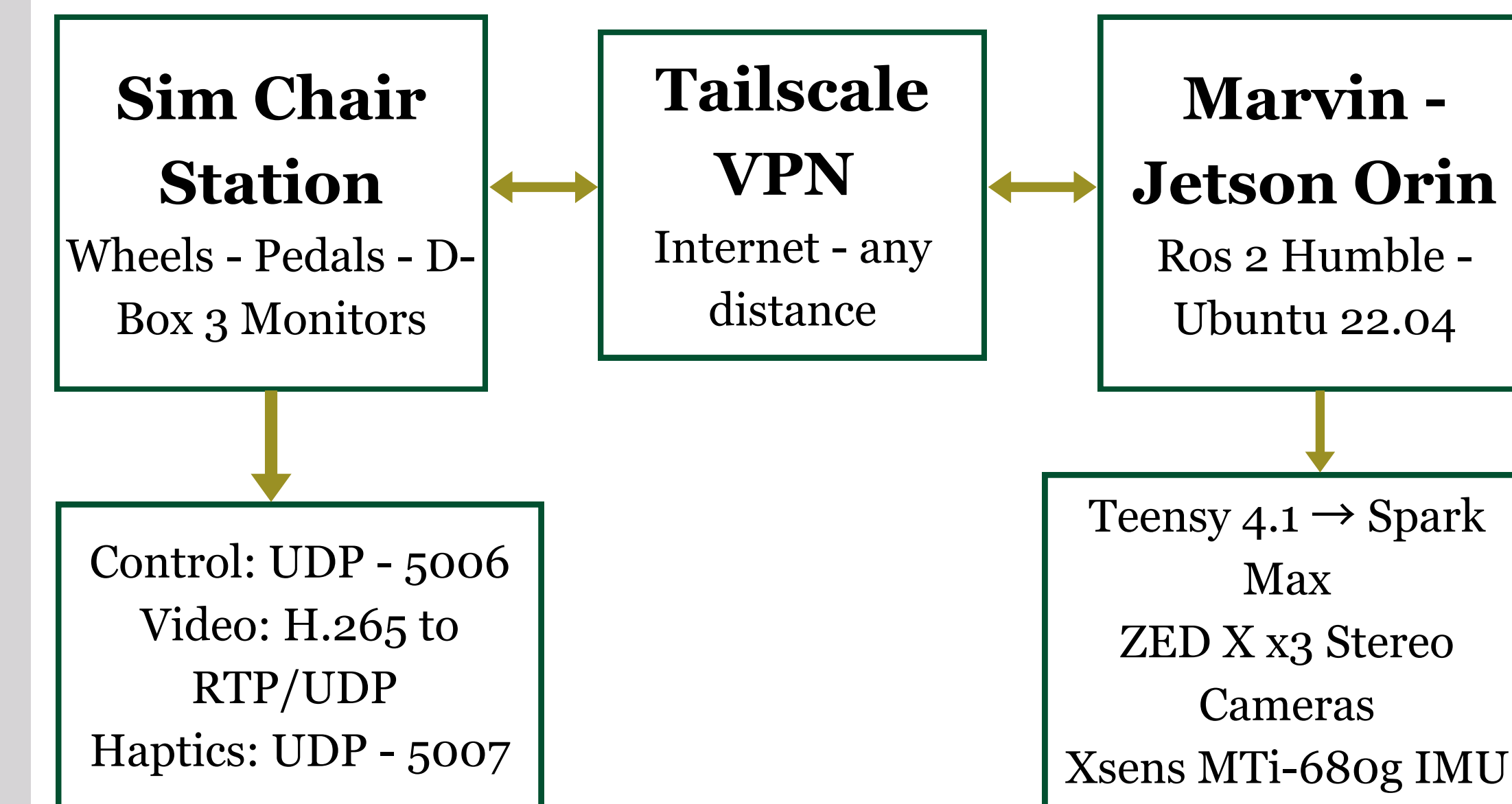


Figure 2. System architecture: control, video, and haptics.

Control (UDP port 5006)

Wheel and pedals send commands to the Jetson, which drives the Teensy and Spark MAX motors. A 300 ms watchdog latches a safe stop if the link drops.

Video (RTP over UDP)

Three ZED X cameras stream H.265 video to three monitors at the chair. Late frames are dropped to keep latency low.

Haptics (UDP port 5007)

The IMU measures kart motion; a washout filter drives the D-BOX chair so the operator feels accel, brake, and cornering. On packet loss, the chair eases to neutral.

Technologies

Jetson Orin · ROS 2 Humble · ZED X (x3) · Xsens MTi-68oG · D-BOX LiveMotion · Tailscale · SDL2 · Python / C++

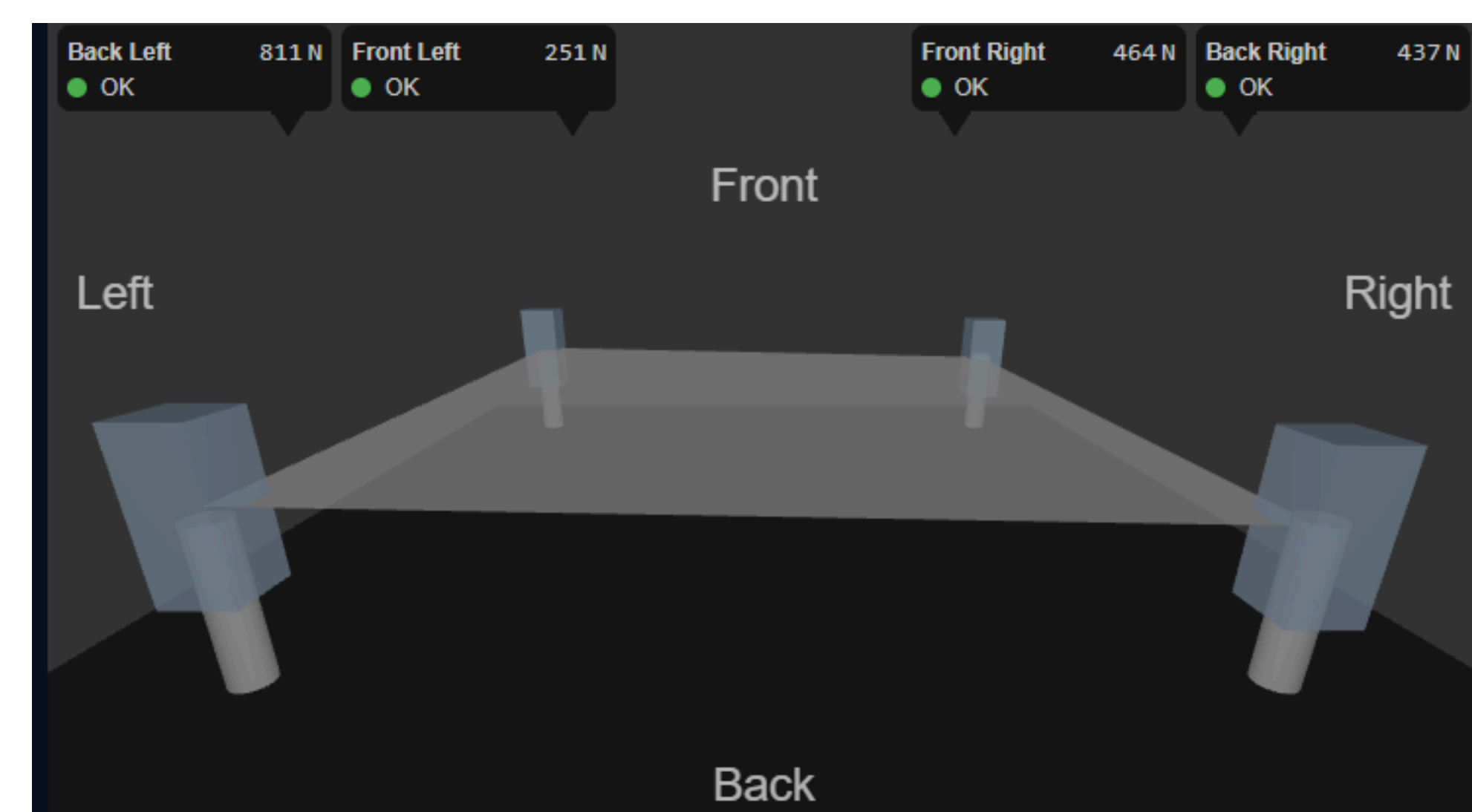


Fig 3: D-Box Sim CHair UI Showing the 4 linear actuator system

Results

Measurements and demonstration status (sim chair to Jetson over Tailscale):

- **Remote** driving: end-to-end operation with live three-camera panoramic video
- **Pipelines:** all three links use UDP (control :5006, video H.265 RTP, haptics :5007)
- **Latency:** network round-trip typically about 95 to 150 ms on the DERP path; direct LAN used when available for lowest latency
- **Video:** ZED X to NVENC H.265 to RTP/UDP to three monitors (late frames dropped for low latency)

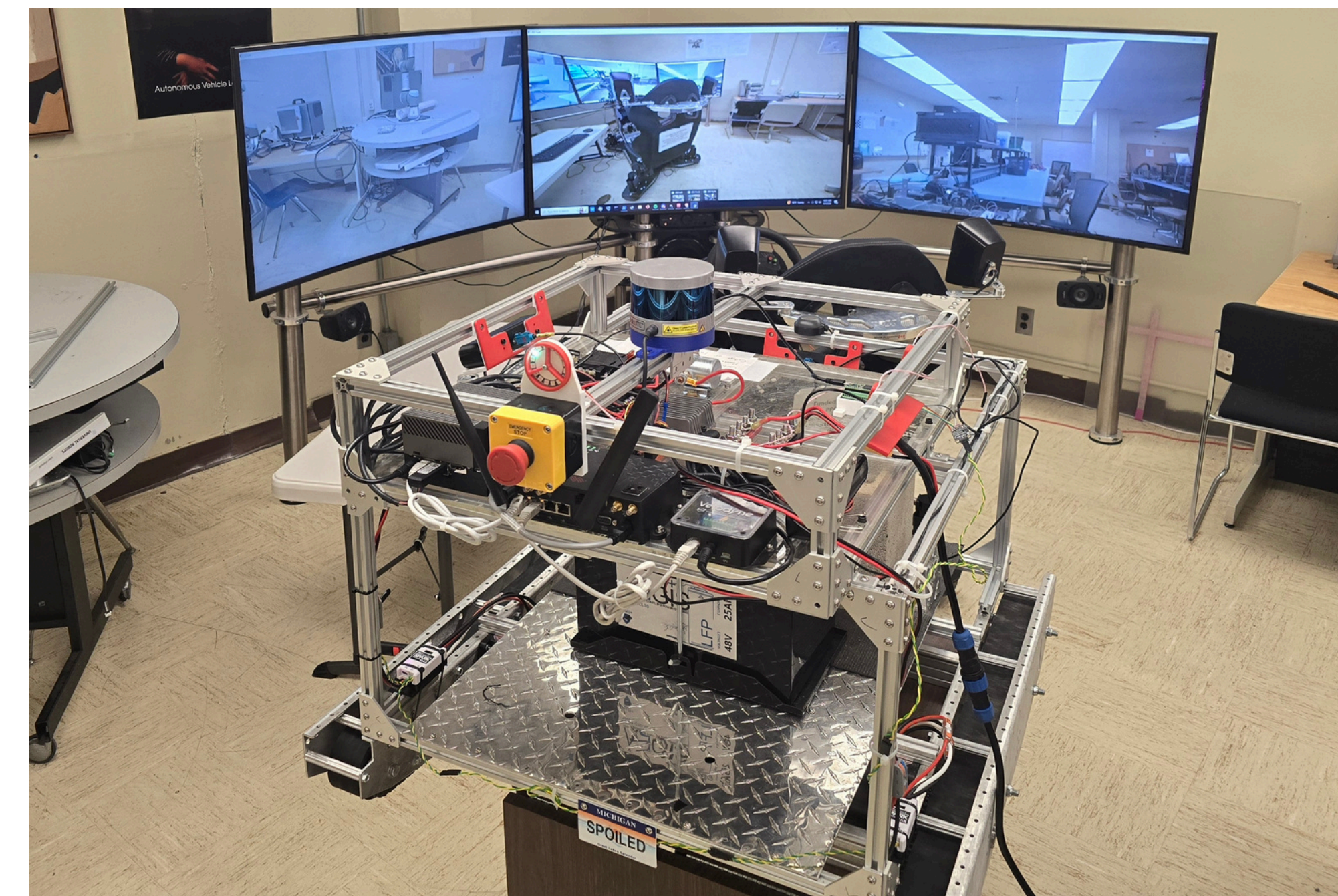


Figure 4. Vehicle-side compute and sensing stack with live three-camera panoramic video on the operator monitors.

- **Failsafe:** control path on UDP :5006 with watchdog-backed stop if the link drops
- **Motion cueing:** haptic path on UDP :5007 (IMU to washout to D-BOX) so the operator feels vehicle motion in the chair
- **Integration:** full simultaneous operation of cameras, wheel/pedals, and motion chair



Figure 5. Operator station: Sparco seat, Thrustmaster wheel and pedals, and D-BOX motion platform.

Together, these measurements show a complete sim-chair to Jetson teleoperation loop over Tailscale: live panoramic video, failsafe control, and motion cueing at the operator station.

Conclusion

This project shows that consumer simulation hardware and an AVL robotics platform can form a full bidirectional teleoperation loop over the internet. Panoramic video, low-latency control, and washout-based motion cueing restore the sensory feedback remote drivers normally lose with a flat camera feed alone.

The same stack supports hazardous-environment operation and autonomy safety intervention, allowing a human to take over with vision and body cues without a line-of-sight radio link. Classical washout on a D-BOX chair, networked video and control, and Tailscale connectivity together give a practical path to immersive remote driving on campus and beyond.

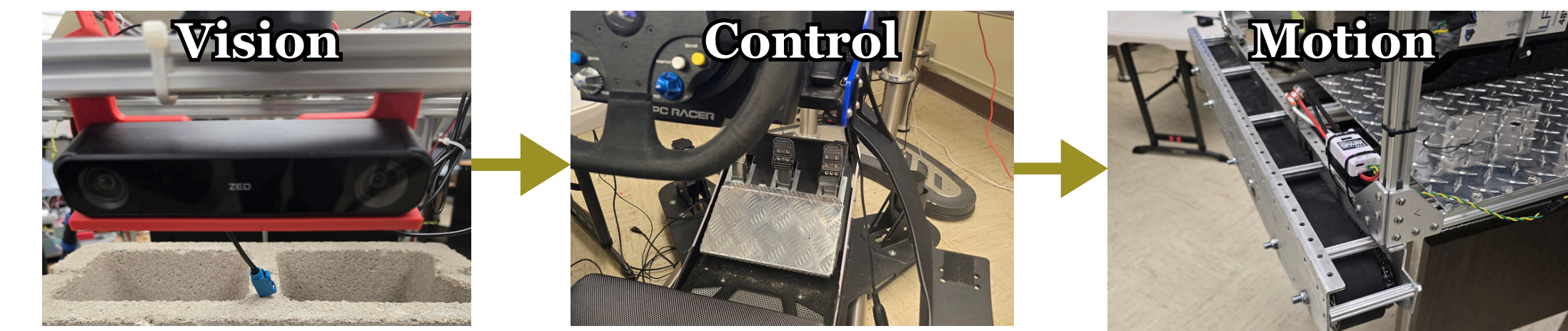


Figure 6. What the operator gets: vision, control, and motion feedback.

Acknowledgements

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References

1. L. Zhao et al., "Remote driving of road vehicles: A survey of driving feedback, latency, support control, and real applications," IEEE Trans. Intell. Vehicles, vol. 9, no. 10, pp. 6086–6107, 2024.
2. L. Zhao, M. Nybacka, and M. Rothhämel, "A survey of teleoperation: Driving feedback," in Proc. IEEE Intell. Vehicles Symp. (IV), 2023, pp. 1–8.
3. M. A. Nahon and L. D. Reid, "Simulator motion-drive algorithms: A designer's perspective," J. Guidance, Control, Dyn., vol. 13, no. 2, pp. 356–362, 1990.
4. P. R. Grant and L. D. Reid, "Motion washout filter tuning: Rules and requirements," J. Aircraft, vol. 34, no. 2, pp. 145–151, 1997.
5. S. B. Kamtam et al., "Network latency in teleoperation of connected and autonomous vehicles: A review of trends, challenges, and mitigation strategies," Sensors, vol. 24, no. 12, Art. no. 3957, 2024.
6. T. Zhang, "Toward automated vehicle teleoperation: Vision, opportunities, and challenges," IEEE Internet Things J., vol. 7, no. 12, pp. 11347–11354, 2020.
7. D. Majstorovic et al., "Survey on teleoperation concepts for automated vehicles," in Proc. IEEE Int. Conf. Syst., Man, Cybern. (SMC), 2022, pp. 1290–1296.
8. L. Zhao et al., "Influence of sound, vibration, and motion-cueing feedback on driving experience and behaviour in real-life teleoperation," in HCI International, Springer, 2024.